

TWILLINGATE, NL, Canada

WMO: 714020

Lat: 49.6880N

Lon: 54.8002W

Elev: 303

StdP: 14.54

Time Zone: -3.50 (NAF)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	4.6	7.8	-3.3	4.6	6.6	-0.4	5.4	9.4	52.3	24.3	45.3	23.5	21.1	280	0.862

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	11.9	76.4	63.8	73.3	62.7	70.3	61.6	66.9	72.6	65.2	70.2	63.4	68.4	11.2	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
64.8	92.8	69.7	63.0	87.2	68.3	61.1	81.4	66.7	31.7	72.5	30.3	70.5	29.0	68.2	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 42.5	36.6	32.0	1.9	83.4	4.3	2.5	-1.1	85.2	-3.6	86.6	-6.0	88.0	-9.1	89.8		
(5)			0.1	70.3	5.2	2.0	-3.7	71.7	-6.8	72.9	-9.7	74.0	-13.5	75.5	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	40.0	22.9	21.4	25.8	33.6	41.2	50.1	60.4	60.8	53.9	44.5	36.2	28.0
	DBStd	15.15	7.12	7.92	6.86	5.89	7.32	8.55	7.08	5.74	6.17	5.91	6.09	6.60
	HDD50	4613	840	801	751	494	293	105	10	2	28	193	415	681
	HDD65	9184	1305	1221	1216	943	739	449	174	154	337	636	864	1146
	CDD50	964	0	0	0	1	20	109	331	335	144	22	2	0
	CDD65	58	0	0	0	0	0	3	30	22	3	0	0	0
	CDH74	214	0	0	0	0	2	26	125	54	7	0	0	0
	CDH80	18	0	0	0	0	0	3	13	2	0	0	0	0
Wind	WSAvg	15.6	19.2	18.1	17.0	15.3	13.6	12.6	11.7	12.0	14.8	16.2	17.5	18.8
Precipitation	PrecAvg	55.9	5.5	5.0	4.6	4.2	3.8	3.8	4.0	3.9	4.8	5.5	5.3	5.6
	PrecMax	66.7	9.2	7.2	8.3	9.4	6.0	5.7	6.4	6.3	8.8	9.5	8.3	9.5
	PrecMin	44.3	3.1	2.6	2.0	1.8	1.6	1.9	2.5	1.9	2.1	1.7	2.4	2.4
	PrecStd	5.4	1.6	1.1	1.3	1.7	1.1	1.0	1.0	1.1	1.6	1.8	1.4	1.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	47.7	46.6	45.9	58.0	69.0	77.3	82.0	78.9	74.2	64.5	57.6	49.4
		MCWB	46.7	44.7	40.5	49.2	54.8	61.5	66.3	66.3	63.8	60.9	55.6	48.2
	2%	DB	40.6	40.1	40.7	51.0	62.7	72.1	77.5	75.2	69.5	59.7	53.7	43.8
		MCWB	39.1	38.9	36.9	45.4	51.7	59.6	64.1	64.0	62.5	55.5	51.8	42.7
	5%	DB	35.7	35.3	37.3	46.2	57.8	67.8	74.1	72.3	66.3	56.3	49.5	39.7
		MCWB	34.4	33.8	34.6	41.5	50.4	57.6	63.1	63.0	60.5	52.8	47.6	38.4
	10%	DB	32.3	31.5	34.3	42.1	53.2	63.4	70.7	69.6	63.2	53.2	44.9	36.6
		MCWB	31.0	30.2	31.7	38.0	47.3	55.8	62.2	61.9	58.0	49.8	42.8	35.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	46.6	45.3	42.8	51.7	58.2	66.0	69.8	69.1	66.4	61.9	55.9	48.1
		MCDB	47.7	46.6	44.5	55.3	65.1	72.1	76.7	75.3	70.7	63.8	57.4	49.3
	2%	WB	39.4	39.0	37.9	46.9	54.6	62.4	67.3	66.7	63.9	56.4	52.1	42.8
		MCDB	40.4	40.1	39.4	50.8	59.5	68.4	73.4	72.0	67.9	58.8	53.5	43.8
	5%	WB	34.5	34.0	34.7	42.3	51.1	59.4	65.4	64.9	61.3	53.2	47.8	38.6
		MCDB	35.6	35.0	37.0	45.6	57.2	65.7	70.8	69.9	65.4	55.7	49.4	39.5
	10%	WB	31.2	30.4	32.1	38.6	47.3	56.5	63.5	63.1	58.6	50.2	43.2	35.5
		MCDB	32.1	31.3	33.8	41.8	53.2	62.6	68.8	68.2	62.5	52.8	44.7	36.6
Mean Daily Temperature Range	5% DB	MDBR	9.3	10.2	9.8	10.0	12.2	13.8	14.3	11.9	10.7	8.2	7.7	7.5
		MCDBR	15.3	16.3	14.6	17.8	22.6	22.2	19.8	16.6	15.6	13.5	15.0	13.0
		MCWBR	15.8	16.0	12.6	13.1	13.8	12.0	9.9	8.8	10.3	11.4	14.8	13.7
	5% WB	MCDBR	15.2	16.4	13.7	17.1	21.3	19.9	17.3	15.3	14.9	13.4	15.1	12.8
		MCWBR	15.7	16.6	12.8	14.2	14.4	12.7	10.9	10.2	11.7	12.5	15.2	13.6
Clear-Sky Solar Irradiance	taub	0.293	0.290	0.314	0.348	0.363	0.379	0.403	0.380	0.356	0.333	0.307	0.302	
	taud	2.348	2.346	2.295	2.327	2.418	2.405	2.405	2.388	2.452	2.504	2.516	2.492	2.307
	Ebn at Noon	239	272	284	285	284	279	270	272	268	255	236	214	
	Edh at Noon	21	27	34	36	35	36	36	32	28	23	19	19	
All-Sky Solar Radiation	RadAvg	276	516	885	1246	1498	1604	1669	1367	1024	621	328	208	
	RadStd	21	70	102	143	134	176	135	95	80	70	34	16	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+450	+405	N/A	N/A
(47) Regional (10 neighbors)	N/A	N/A	+2.45	N/A	+0.81	+1.13	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page